

AI-BASED SMART GREENHOUSE AUTOMATION PLATFORM



KNOWLEDGE INSTITUTE OF TECHNOLOGY

AI-BASED SMART GREENHOUSE AUTOMATION PLATFORM

Smart IoT-Based Agriculture Monitoring, Automation & AI Suggestion System

Main Technologies:

- ESP32
- Blynk IoT
- DHT22 Temperature & Humidity Sensor
- Soil Moisture Sensor
- Automatic Irrigation & Ventilation
- Visual Studio Code
- Python AI / Machine Learning

Double Content Slide

- **AI-Based Smart Greenhouse Automation Platform**
- An IoT-based system that monitors greenhouse environmental conditions and automatically controls the required devices for better crop growth.
- The **ESP32** collects temperature, humidity, soil moisture and light data.
- The data is displayed locally and remotely through **Blynk IoT**, while **Visual Studio Code with Python AI** analyzes the readings and provides intelligent suggestions.

OBJECTIVES

- To develop a **compact, low-cost and intelligent greenhouse automation system** Monitor temperature and humidity.
- Measure soil moisture.
- Monitor light intensity.
- Automatically control water pump and fan.
- Provide remote monitoring using **Blynk IoT**.
- Analyze sensor readings using **Visual Studio Code + Python AI**.
- Provide AI-based suggestions for greenhouse management.

Bulleted List Slide

- **DHT22** – Measures temperature and humidity
- **Soil Moisture Sensor** – Measures soil moisture
- **LDR / Light Sensor** – Measures light intensity
- **ESP32** – Main controller and Wi-Fi communication
- **Relay Module** – Controls greenhouse devices
- **Water Pump** – Automatic irrigation
- **Exhaust Fan** – Temperature and humidity control
- **LED Grow Light** – Provides additional lighting
- **Blynk IoT** – Remote monitoring
- **Visual Studio Code** – AI application development
- **Python AI/ML** – Sensor-data analysis and suggestions



Comparison Slide

COMPARE SECTION

- Traditional greenhouse systems mainly depend on manual monitoring and control of irrigation, temperature and lighting.
- The proposed system uses **ESP32 and sensors** to continuously monitor greenhouse conditions and automatically control the required devices.

CONTRAST SECTION

- Unlike conventional greenhouse monitoring, this system integrates:
- **ESP32**
- **Blynk IoT**
- **Visual Studio Code**
- **Python AI**
- **Automatic control**
- The farmer can remotely monitor greenhouse conditions through **Blynk**, while the AI analyzes readings and provides suggestions.

Prototype

- The prototype consists of an **ESP32** connected to **temperature, humidity, soil moisture and light sensors**.
- The ESP32 collects the sensor readings and controls the **water pump, exhaust fan and grow light through relay modules**.
- The sensor data is sent through Wi-Fi to the **Blynk IoT dashboard** for remote monitoring.
- The same readings are processed in **Visual Studio Code using Python AI**, which generates suggestions for greenhouse management.



Detection Breakdown

 AI-BASED SMART GREENHOUSE AUTOMATION PLATFORM 					
LIVE READINGS & AI SUGGESTIONS					
PARAMETER	SENSOR READING	CONDITION	AI SUGGESTION	AUTOMATIC ACTION	STATUS
 Temperature	34 °C	High	Temperature is high	Fan ON	 ON
 Humidity	42 %	Low	Humidity is low	Misting ON	 ON
 Soil Moisture	28 %	Dry	Soil is dry	Water Pump ON	 ON
 Light Intensity	25 %	Low	Light is insufficient	Grow Light ON	 ON
 Temperature	28 °C	Normal	Temperature is suitable	Fan OFF	 OFF
 Humidity	65 %	Normal	Humidity is suitable	Misting OFF	 OFF
 Soil Moisture	55 %	Normal	Soil moisture is optimal	Water Pump OFF	 OFF
 Light Intensity	70 %	Sufficient	Light level is sufficient	Grow Light OFF	 OFF

EXAMPLE : LIVE READING → AI ANALYSIS → SUGGESTED ACTION		
LIVE READING	AI ANALYSIS	SUGGESTED ACTION
 Temperature = 34 °C	Temperature is high	 Turn ON Exhaust Fan
 Humidity = 42 %	Humidity is low	 Turn ON Misting System
 Soil Moisture = 28 %	Soil is dry	 Turn ON Water Pump
 Light = 25 %	Light is low	 Turn ON Grow Light

NOTE

These are example/project threshold readings for demonstrating the system. Actual thresholds should be calibrated according to the crop and greenhouse conditions.



CONCLUSION

- The **AI-Based Smart Greenhouse Automation Platform** combines **ESP32**, **sensors**, **automatic control**, **Blynk IoT** and **Visual Studio Code** with **Python AI**.
- The system provides **real-time monitoring**, **remote access**, **automatic control** and **AI-based suggestions** for efficient greenhouse management.
- It provides a practical foundation for **smart agriculture**, **reduced manual intervention** and **efficient resource utilization**.